

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log

The UDP protocol reveals that the DNS server is not responding to requests for the domain yummyrecipesforme.com.

This is based on the results of the network analysis, which show that when the client (192.51.100.15) sent a DNS query to 203.0.113.2, the reply that came back was not a DNS answer at all, but an ICMP message stating “udp port 53 unreachable.”

The port noted in the error message, port 53, is used for DNS protocol traffic.

The most likely issue is that the DNS server itself is down, or something is actively blocking UDP traffic to port 53 on that server.

Part 2: Explain your analysis of the data and provide at least one cause of the incident

The incident was first observed at 13:24 in the afternoon, when the client machine attempted to resolve yummyrecipesforme.com and received a “destination port unreachable” error in the browser.

To investigate, the network security team loaded tcpdump and repeated the request to capture what was actually happening on the wire. The capture showed three separate DNS query attempts, at 13:24, 13:26, and 13:28, and each one was met with the same ICMP error from the DNS server: port 53 unreachable.

Because this happened consistently across multiple attempts spaced several minutes apart, rather than as a one-time glitch, it points to a persistent problem rather than a brief network hiccup.

The next step for the team is to check whether the DNS server itself has crashed or is misconfigured, or whether a firewall rule is blocking inbound UDP port 53 traffic to it.

A likely cause is that the DNS service on that server went down — possibly due to a crash, a failed update, or even a denial-of-service attempt aimed at the DNS server — though at this stage a firewall misconfiguration blocking the port cannot be ruled out either.